RWANDA POLYTECHNIC
KIGALI COLLEGE

IAP FINAL REPORT

Prepared by:

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ICT DEPARTMENT
IT PROGRAM
LEVEL 7

pryro

DECLARATION

I, **UMURERWA Rosine**, hereby declare that this Industrial Attachment Report is my original work and has been prepared based on the knowledge, skills, and practical experience gained during my industrial attachment. The information presented in this report is true and accurate to the best of my knowledge and has not been submitted to any other institution for academic assessment or any other purpose.

UMURERWA Rosine

21RP04054

Signature:

Date: 28/06/2026

DEDICATION

I dedicate this report to my beloved family for their endless love, encouragement, and unwavering support throughout my academic journey. Their sacrifices, prayers, and motivation have been a constant source of strength and inspiration.

I also dedicate this work to my lecturers, supervisors, and mentors whose guidance, knowledge, and encouragement have contributed significantly to my personal and professional growth.

Finally, I dedicate this report to all my fellow students and future Information Technology professionals, with the hope that it serves as an inspiration to embrace continuous learning, innovation, and excellence in the field of technology.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Rwanda Polytechnic Kigali College** for providing me with the opportunity to undertake my industrial attachment and apply the knowledge and skills acquired during my studies in a real working environment.

I also extend my heartfelt appreciation to the management and staff of **Pryro INC** for warmly welcoming me, providing a supportive working environment, and giving me the opportunity to participate in various software development projects throughout my attachment period.

My special thanks go to my industrial supervisor and all my colleagues for their continuous guidance, encouragement, and willingness to share their knowledge and experience. Their support greatly contributed to my professional growth and the successful completion of my attachment.

Finally, I thank my family, friends, and everyone who supported and encouraged me throughout this journey. Their motivation and support played a significant role in helping me successfully complete my industrial attachment.

ABSTRACT

This report presents the activities, experiences, and knowledge gained during my Industrial Attachment, which was undertaken as part of the requirements for the award of a Diploma in Information Technology at Rwanda Polytechnic Kigali College. The attachment provided an opportunity to apply theoretical knowledge in a real working environment while developing practical and professional skills.

During the attachment, I participated in various software development activities, with a major focus on the design and development of **CareerMind AI**, an AI-powered career guidance platform that assists users with career planning, CV and cover letter generation, interview preparation, and personalized career recommendations. In addition, I was involved in UI/UX design, inventory management systems, healthcare software research, graphic design, multimedia content creation, and communication skills development.

The project was developed using modern web technologies, including **React.js**, **Next.js**, **Node.js**, **Supabase**, and other supporting tools. These technologies enabled the development of responsive, secure, and user-friendly applications while strengthening my understanding of full-stack web development.

The industrial attachment significantly enhanced my technical competencies, problem-solving abilities, teamwork, communication, and time management skills. Overall, the experience bridged the gap between classroom learning and real-world practice, preparing me for future professional responsibilities in the field of Information Technology.

Contents

CHAPTER ONE: INTRODUCTION

1.1. Historical Background

Pryro is an ERP platform company based in Kigali, Rwanda. Founded in 2020, Pryro was born from a vision to solve fragmented business systems. Our experienced entrepreneur founders understood the pain of juggling multiple platforms for ERP, HRM, CRM, accounting, and project management. Today, Pryro is trusted by 64,000+ businesses across 5 continents, from startups to enterprises

1.2. Organizational Profile

Here's a brief overview of pryro's organizational profile:

- Location: 1 KN 78 Nyarugenge Street, Kigali, Rwanda
- Contact: +250 788 715 075, sales@pryro.com, support@pryro.com
- Founded: 2020 * Customers: 64,000+ businesses across 5 continents
- Solution: Complete ERP solution with AI-powered insights to manage finance, inventory, HR, and operations in one unified platform.

It is a global company, serving businesses worldwide with our intelligent ERP solutions

1.3. Vision and Mission

Vision

Pryro's vision is to solve fragmented business systems, providing a unified platform for businesses to manage their operations, and empowering them to make data-driven decisions with the help of AI-powered insights. Our vision is to be the go-to ERP solution for businesses worldwide, helping them to optimize their processes, increase efficiency, and drive growth.

Mission

Pryro's mission is to provide a complete ERP solution that empowers businesse s to streamline their operations effortlessly, with AI-powered insights. We aim to help businesses manage their finance, inventory, HR, and operations in one unified platform, making it easier for them to achieve their goals and scale their business.

1.4. The core values

✧ **Mission-Driven**

We exist to empower businesses through innovative technology solutions

✧ **Customer-First**

Your success is our success. We listen, adapt, and deliver

✧ **Innovation**

Constantly pushing boundaries to stay ahead of tomorrow's challenges

✧ **Security & Trust**

Enterprise-grade protection with complete transparency

1.5. Organizational structure

1. **Management Team**: Oversees overall strategy and direction.
2. **Sales Team**: Handles sales and business development.
3. **Support Team**: Provides customer support and assistance.
4. **Development Team**: Responsible for product development and maintenance.
5. **Operations Team**: Manages day-to-day operations, including HR, finance, and administration.

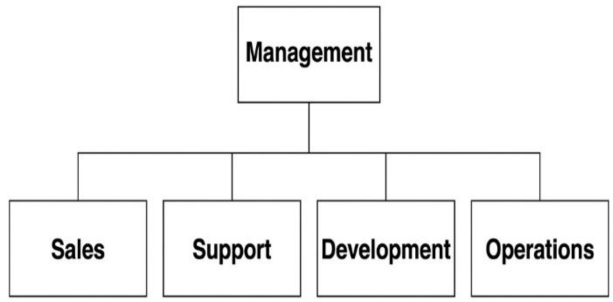


Figure 1: organizational structure

Conclusion

This chapter introduced Pryro by presenting its historical background, organizational profile, vision, mission, core values, and organizational structure. Understanding the company's operations and objectives provided a solid foundation for my industrial attachment and helped me appreciate the professional environment in which I developed my technical and professional skills.

CHAPTER 2 : PERFORMED ACTIVITIES IN INDUSTRIAL ATTACHMENT

2.1 General Activities Undertaken

During my industrial attachment, I participated in a variety of technical and professional activities that enhanced my knowledge and practical skills in software development and information technology. The major activities undertaken include:

1. **Company Orientation**
    - Learned about the company's working environment, policies, objectives, and organizational structure.
2. **Software Development**
    - Participated in the design and development of web applications using modern technologies such as React.js, Next.js, Node.js, and Supabase.
3. **CareerMind AI Development**
    - Contributed to the development of CareerMind AI, an AI-powered career guidance platform. This involved designing user interfaces, implementing system features, and integrating backend services.
4. **UI/UX Design**
    - Applied user interface and user experience design principles to create responsive, attractive, and user-friendly application interfaces.
5. **Inventory Management System**
    - Learned how inventory management systems operate, including stock creation, stock management, and updating system features.
6. **Healthcare Software Research**
    - Studied healthcare software development, including system design standards, privacy policies, terms and conditions, and regulatory requirements.
7. **Graphic Design**
    - Designed logos, flyers, and promotional materials using Canva while applying basic graphic design principles.
8. **Multimedia and Social Media Advertising**
    - Learned how to create promotional videos and social media advertisements for digital marketing purposes.
9. **Communication and Professional Development**
    - Improved communication, teamwork, problem-solving, time management, and workplace professionalism through daily collaboration and project activities.

2.2 Specific Activities Undertaken

2.2.1 CareerMind AI Development

Activities Performed

- Participated in the design and development of CareerMind AI.
- Implemented system features and improved existing modules.
- Tested the application and fixed software bugs.
- Collaborated in planning and improving the system.

Skills Gained

- AI application development.
- Software development lifecycle.
- System testing and debugging.
- Problem-solving and teamwork.

Tools Used

- React.js
- Next.js
- Node.js
- Supabase
- Kiro Code Editor

2.2.2 Frontend Development

Activities Performed

- Designed responsive web pages.
- Built reusable React components.
- Developed authentication, dashboard, and profile pages.
- Improved application responsiveness.

Skills Gained

- Frontend development.
- Responsive web design.
- Component-based development.
- UI implementation.

Tools Used

- React.js
- Next.js
- HTML

- Tailwind CSS
- Type script
- JavaScript

2.2.3 Backend Development

Activities Performed

- Developed backend logic.
- Implemented REST APIs.
- Processed user requests.
- Connected frontend with the database.

Skills Gained

- Backend programming.
- API development.
- Server-side logic.
- Data processing.

Tools Used

- Node.js
- JavaScript
- Supabase

2.2.4 Database Management Using Supabase

Activities Performed

- Created and managed database tables.
- Managed user authentication.
- Stored and retrieved application data.
- Maintained data consistency.

Skills Gained

- Database management.
- Authentication.
- CRUD operations.
- Cloud database administration.

Tools Used

- Supabase

- SQL
- Node.js

2.2.5 User Interface and User Experience (UI/UX) Design

Activities Performed

- Designed user-friendly interfaces.
- Improved layouts and navigation.
- Applied UI/UX design principles.
- Enhanced system usability.

Skills Gained

- UI Design.
- UX Design.
- Design thinking.
- User-centered development.

Tools Used

- Figma
- React.js
- Tailwind CSS

2.2.6 Inventory Management System

Activities Performed

- Learned inventory management processes.
- Created and updated stock records.
- Managed inventory information.
- Explored inventory system features.

Skills Gained

- Inventory management.
- Stock control.
- Business system understanding.
- Data management.

Tools Used

- Inventory Management System
- Supabase

- Kiro Code Editor

2.2.7 Healthcare Software Research

Activities Performed

- Researched healthcare software.
- Studied software policies and regulations.
- Analyzed privacy and security requirements.
- Reviewed healthcare system standards.

Skills Gained

- Research skills.
- Software compliance.
- Documentation.
- Critical thinking.

Tools Used

- Internet
- Research articles
- Documentation

2.2.8 Graphic Design

Activities Performed

- Designed logos.
- Created flyers.
- Produced marketing materials.
- Applied graphic design principles.

Skills Gained

- Graphic design.
- Branding.
- Creativity.
- Visual communication.

Tools Used

- Canva

2.2.9 Multimedia and Social Media Advertising

Activities Performed

- Designed social media advertisements.
- Created promotional content.
- Learned basic video editing.
- Planned digital marketing materials.

Skills Gained

- Digital marketing.
- Content creation.
- Multimedia production.
- Advertising techniques.

Tools Used

- Social media platforms
- YouTube Videos for learning

2.2.10 Communication and Professional Development

Activities Performed

- Participated in team meetings.
- Collaborated with colleagues.
- Presented project ideas.
- Managed assigned tasks.

Skills Gained

- Communication.
- Teamwork.
- Time management.
- Professional ethics.
- Adaptability.
- Problem-solving.

Tools Used

- Email
- Visual communication

Conclusion

This chapter presented the general and specific activities carried out during my industrial attachment. Through these activities, I gained practical experience in software development, AI, UI/UX design, database management, and professional communication. Overall, the attachment enhanced my technical skills, problem-solving abilities, and preparedness for a career in Information Technology.

CHAPTER THREE: INDUSTRIAL ATTACHMENT LIFE EXPERIENCE

3.1 Introduction

Industrial attachment provided me with an excellent opportunity to experience the professional working environment and apply the theoretical knowledge acquired during my academic studies. It enabled me to understand how software development projects are planned, developed, tested, and maintained in a real organization. Throughout the attachment period, I worked with experienced professionals who guided me in performing different technical and professional tasks while helping me improve my confidence and practical abilities.

3.2 Working Environment

The working environment was professional, collaborative, and supportive. Team members were approachable and willing to provide guidance whenever I encountered challenges. The organization promoted teamwork, innovation, continuous learning, and knowledge sharing, making it easier for me to adapt to workplace culture and contribute to ongoing projects.

I was provided with the necessary tools and resources to perform my assigned duties, including access to development environments, project documentation, and learning materials. This enabled me to complete my tasks efficiently while improving my technical knowledge.

3.3 Relationship with Supervisors and Colleagues

Throughout the attachment period, I loved working with my supervisors and colleagues. They provided continuous guidance, constructive feedback, and technical support whenever needed. Regular discussions and collaborative problem-solving sessions helped me better understand software development practices and workplace expectations.

Working with professionals also improved my communication skills, confidence, teamwork, and ability to receive and apply constructive criticism.

3.4 Skills and Knowledge Acquired

The industrial attachment significantly enhanced both my technical and professional competencies.

Technical Skills

During the attachment, I improved my knowledge of full-stack web development using React.js, Next.js, Node.js, and Supabase. I gained practical experience in frontend and backend development, database management, UI/UX design, software testing, debugging, and AI application development through the CareerMind AI project.

Professional Skills

The attachment also strengthened my communication, teamwork, time management, adaptability, critical thinking, and problem-solving skills. Working in a professional environment taught me how to prioritize tasks, meet deadlines, and maintain high standards of professionalism.

3.5 Challenges Encountered and Solutions

Like any learning experience, the industrial attachment presented several challenges. Initially, adapting to new technologies and understanding existing project structures(the purpose of project) required additional effort. I addressed these challenges by reading documentation, conducting online research, asking questions, and seeking guidance from experienced colleagues.

Another challenge involved debugging application errors and integrating different technologies. I overcame these difficulties through continuous testing, teamwork, and systematic troubleshooting until appropriate solutions were found.

3.6 Lessons Learned

The industrial attachment taught me that continuous learning is essential in the field of Information Technology because technologies evolve rapidly. I learned the importance of planning before development, writing clean and maintainable code, testing software regularly, and documenting project progress.

The experience also emphasized the value of teamwork, professionalism, effective communication, responsibility, and commitment in achieving successful project outcomes.

3.7 Overall Experience

Overall, my industrial attachment was a highly rewarding and educational experience. It allowed me to bridge the gap between theoretical knowledge and practical application while developing both technical expertise and professional confidence.

Working on projects such as CareerMind AI exposed me to modern software development practices and strengthened my passion for software engineering and artificial intelligence. The experience has prepared me to face future professional challenges with confidence and has motivated me to continue learning and improving my skills throughout my career.

CHAPTER FOUR: CareerMind AI

4.1 Introduction

This chapter presents the case study conducted during the industrial attachment, which focused on the design and development of **CareerMind AI**, an artificial intelligence-powered career guidance platform. The case study was based on students at **Rwanda Polytechnic Kigali College**, with the aim of understanding the challenges they face in career planning and demonstrating how technology can be used to address those challenges.

The chapter describes the background of the study, the problem identified, the justification for developing the system, the objectives of the project, and the scope of the study. It lays the foundation for the design and implementation of CareerMind AI as an innovative solution for career development.

4.2 Background of the Case Study

Choosing a career and preparing for employment are among the most important decisions students make during their academic journey. However, many students face difficulties when selecting a suitable career path, preparing professional resumes and cover letters, identifying the skills required by employers, and preparing for job interviews. These challenges are often caused by limited access to career guidance services, insufficient practical experience, and a lack of personalized support.

At **Rwanda Polytechnic Kigali College**, students are trained in different technical and professional fields that prepare them for the job market. Despite this, many students still require additional guidance to connect their academic knowledge with career opportunities. Traditional career guidance methods, such as occasional workshops or counseling sessions, may not always provide continuous, personalized support for every student.

With the rapid advancement of Artificial Intelligence (AI), intelligent systems can now provide personalized recommendations, automate repetitive tasks, and deliver career support at any time. AI-powered platforms have the potential to analyze user information, recommend suitable career paths, assist in preparing application documents, and help users improve their employability.

To address these challenges, **CareerMind AI** was developed as an intelligent career guidance platform designed to support students throughout their career development journey. The platform integrates AI-driven tools to help users create professional resumes and cover letters, prepare for interviews, receive personalized career recommendations, and access career-related resources through a modern and user-friendly web application.

4.3 Problem Statement

Many students at Rwanda Polytechnic Kigali College experience challenges in making informed career decisions and preparing for employment after graduation. Although they acquire technical knowledge during their studies, they often lack personalized guidance on career planning, resume writing, interview preparation, and professional development.

Most students rely on internet searches, social media, or general online resources that provide generic information rather than advice tailored to their individual skills, interests, and career goals. In addition, access to career guidance services may be limited by time, availability, or the number of students seeking assistance.

As a result, students may struggle to identify suitable career opportunities, produce competitive job application documents, or confidently prepare for interviews. These challenges can reduce their readiness for the labor market and affect their ability to secure employment.

Therefore, there is a need for an intelligent and accessible platform that provides personalized career guidance and supports students in preparing for successful careers.

4.4 Justification of the Study

The development of CareerMind AI was motivated by the need to improve career guidance services for students through the use of Artificial Intelligence. The system was designed to provide continuous, personalized, and easily accessible career support that complements existing career guidance initiatives.

The project is significant because it enables students to receive career advice at any time, generate professional resumes and cover letters more efficiently, prepare for interviews using AI-assisted tools, and gain confidence before entering the job market.

From an educational perspective, the project demonstrates how modern technologies such as Artificial Intelligence, full-stack web development, and cloud databases can be applied to solve real-world challenges within higher education institutions.

The study also provided an opportunity to apply theoretical knowledge gained during academic studies to the design and implementation of a practical software solution, thereby enhancing technical skills and professional competence.

4.5 Objectives of the Study

4.5.1 General Objective

To design and develop **CareerMind AI**, an AI-powered career guidance platform that assists all students and all people who's in career planning, professional document preparation, and employment readiness.

4.5.2 Specific Objectives

The specific objectives of the project were to:

- Analyze the career guidance challenges faced by students at Rwanda Polytechnic Kigali College.
- Design a user-friendly AI-powered career guidance platform.
- Develop features for career recommendations, resume generation, cover letter creation, and interview preparation.
- Implement secure user authentication and database management.
- Improve access to career guidance services through modern web technologies.
- Evaluate the effectiveness of the system in supporting career development.

4.6 Scope of the Study

This case study focuses on the design, development, and implementation of CareerMind AI as a career guidance platform for students at Rwanda Polytechnic Kigali College.

The study covers the analysis of career-related challenges experienced by students, the design of system features, the implementation of the web application using modern technologies, and the evaluation of its potential benefits.

The system provides functionalities such as user registration and authentication, AI-powered career guidance, resume and cover letter generation, interview preparation support, personalized career recommendations, and a user dashboard for managing career development activities.

The scope of the study is limited to students of Rwanda Polytechnic Kigali College as the primary users of the system. Although the platform has the potential to support graduates and job seekers from other institutions, these groups are outside the scope of this case study.

4.7 Requirements Analysis

Requirements analysis is one of the most important stages in software development because it helps identify the needs and expectations of users before the system is designed and implemented. During the development of CareerMind AI, the requirements were gathered through observation of common challenges faced by students at Rwanda Polytechnic Kigali

College, discussions with fellow students, research on career guidance practices, and analysis of existing career development platforms.

The analysis revealed that many students need a centralized platform that provides personalized career guidance, assists in preparing professional documents, and helps them improve their employability. Based on these findings, both functional and non-functional requirements were identified to guide the development of the system.

4.8 Functional Requirements

Functional requirements describe the services and operations that the system must perform to meet user needs. CareerMind AI was designed with the following key functionalities:

User Registration and Authentication

The system allows users to create an account, log in securely, and manage their personal profiles. Authentication ensures that only authorized users can access their information.

AI Career Guidance

Users can receive AI-powered career recommendations based on their educational background, skills, interests, and career goals.

Resume (CV) Generator

The system enables users to generate professional resumes by entering their personal details, education, skills, work experience, and achievements. The generated resume can be reviewed and downloaded for job applications.

Cover Letter Generator

CareerMind AI assists users in creating personalized cover letters that match specific job opportunities and professional requirements.

Interview Preparation

The platform provides interview questions, preparation tips, and AI-generated feedback to help users improve their confidence before interviews.

User Dashboard

Each registered user has access to a personal dashboard where they can manage their career information, access AI tools, and monitor their progress.

Subscription Management

The system supports subscription plans that allow users to access premium AI-powered career services while maintaining free basic functionality.

Administration

The administrator can manage users, monitor system activities, update content, and ensure the platform operates efficiently.

4.9 Non-Functional Requirements

Non-functional requirements define the quality attributes that determine how well the system performs.

Performance

The system should respond quickly to user requests and process AI-generated results within an acceptable time.

Security

User information should be protected through secure authentication mechanisms and proper database access control. Personal career information must remain confidential.

Reliability

The platform should be available whenever users need career assistance and should operate consistently without frequent failures.

Usability

The interface should be simple, attractive, and easy to use for students with different levels of computer literacy.

Scalability

The system should be capable of supporting an increasing number of users and additional AI features in the future without significant performance degradation.

Maintainability

The application should be structured in a way that allows developers to update features, fix bugs, and improve the system efficiently.

Compatibility

CareerMind AI should function correctly on different web browsers and devices, including desktop computers, tablets, and smartphones.

4.10 Feasibility Study

Before developing CareerMind AI, a feasibility study was conducted to determine whether the project could be successfully implemented.

Technical Feasibility

The project was technically feasible because modern technologies such as React.js, Next.js, Node.js, and Supabase provide the necessary tools for developing a secure, scalable, and responsive web application. The availability of AI technologies further supported the implementation of intelligent career guidance features.

Economic Feasibility

The project required relatively low development costs because most of the technologies and development tools used are open source or provide free development plans. This reduced implementation expenses while maintaining high-quality system performance.

Operational Feasibility

Students frequently require career guidance throughout their academic journey, making the system practical and useful within an educational environment. The web-based nature of CareerMind AI allows users to access career support at any time using an internet-connected device.

Schedule Feasibility

The project was completed within the available industrial attachment period through proper planning, task prioritization, and continuous development and testing.

4.11 System Design

System design translated the identified requirements into a structured software solution capable of meeting user needs efficiently. During this phase, the overall architecture, database structure, user interface, and navigation flow were designed before implementation.

The design process focused on developing a modern, responsive, and user-friendly application that would provide students with an intuitive experience while accessing career guidance services.

The major system modules include:

- User Authentication Module
- User Profile Management Module
- AI Career Guidance Module
- Resume Generator Module
- Cover Letter Generator Module
- Interview Preparation Module
- Subscription Management Module
- Administrative Dashboard

The relationship between these modules enables users to move seamlessly through the application while maintaining secure access to their personal information.

Figure 4.1: Overall System Architecture

The modular design approach adopted during development improves maintainability, scalability, and future expansion of the system. Additional features can be integrated into the platform without affecting existing functionalities, making CareerMind AI flexible for future improvements.

4.12 System Architecture

The architecture of CareerMind AI was designed using a modern web application architecture that separates the presentation layer, application logic, and data management. This modular approach improves scalability, maintainability, and security while ensuring efficient communication between different system components.

The system follows a client-server architecture where users interact with the frontend application through a web browser. User requests are processed by the backend server, which communicates with the database and AI services before returning the requested information to the user.

The main components of the system include:

- **Frontend Layer:** Developed using React.js and Next.js to provide a responsive and interactive user interface.
- **Backend Layer:** Implemented using Node.js to process business logic, manage user requests, and integrate different services.
- **Database Layer:** Managed using Supabase to store user information, authentication data, resumes, career profiles, and application records.
- **Artificial Intelligence Layer:** Provides intelligent career recommendations, resume assistance, cover letter generation, and interview preparation support.

- **Authentication Module:** Ensures secure login, registration, and user account management.

This architecture enables efficient communication between users and the system while ensuring data security and reliable performance.

4.13 Database Design

The database plays an essential role in storing and managing information required by the system. CareerMind AI uses **Supabase**, a cloud-based PostgreSQL database that provides secure data storage, authentication, and real-time database services.

The database was designed to maintain consistency, reduce redundancy, and support efficient retrieval of user information.

The main database entities include:

- Users
- User Profiles
- Career Assessments
- Resumes
- Cover Letters
- Interview Sessions
- Subscription Plans
- Payments
- Career Recommendations

These entities are related through primary and foreign keys, ensuring that information remains organized and consistent.

The database supports CRUD (Create, Read, Update, Delete) operations, allowing users to manage their career information efficiently.

4.14 Technologies Used

The development of CareerMind AI involved several modern technologies that supported frontend development, backend implementation, database management, and AI integration.

Technology	Purpose
React.js	Building interactive user interfaces
Next.js	Developing responsive web applications
Node.js	Backend development and API implementation

Technology	Purpose
Supabase	Database management and user authentication
JavaScript	Programming language used throughout the application
HTML5	Structuring web pages
CSS3 / Tailwind CSS	Styling and responsive layouts
Git & GitHub	Version control and collaboration
Kiro Code Editor	Writing and managing source code
AI APIs	Providing intelligent career guidance and content generation

These technologies were selected because they are modern, scalable, reliable, and suitable for full-stack web application development.

4.15 System Implementation

After completing the planning and design phases, the implementation phase involved transforming the system requirements into a fully functional web application.

Development began with setting up the project structure and configuring the frontend and backend environments. Individual modules were then developed independently before being integrated into a complete system.

The implementation process included:

- Designing responsive web interfaces.
- Developing reusable frontend components.
- Creating backend APIs.
- Integrating the frontend with Supabase.
- Implementing secure user authentication.
- Connecting AI services to provide career recommendations.
- Performing software testing and debugging.

Each module was tested individually before integration to ensure that it performed its intended function correctly.

4.16 User Interface Implementation

One of the primary goals of CareerMind AI was to provide a simple, attractive, and user-friendly interface.

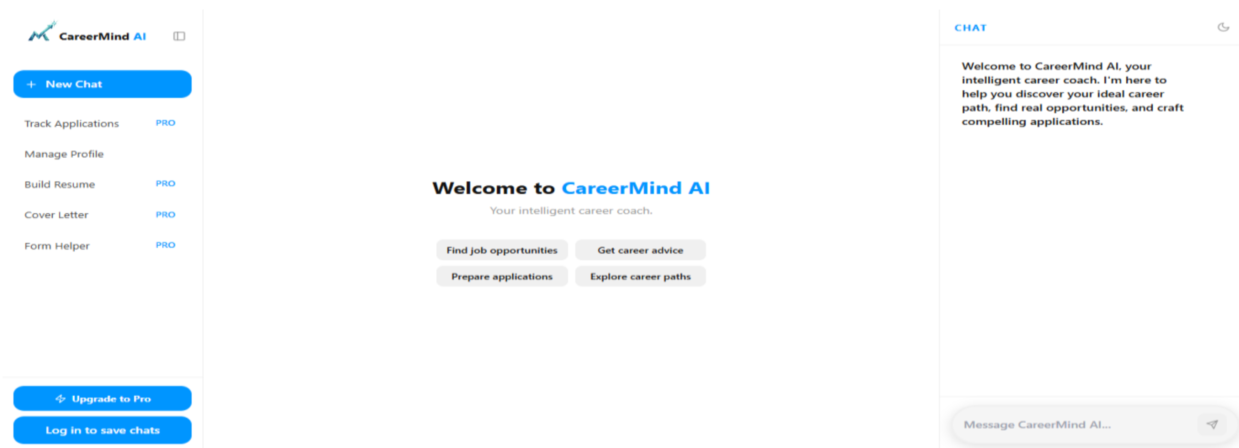
The interface was designed to minimize complexity while allowing users to access career guidance services quickly and efficiently.

The main system interfaces include:

Home Page

The home page introduces users to CareerMind AI and provides navigation to different services available on the platform.

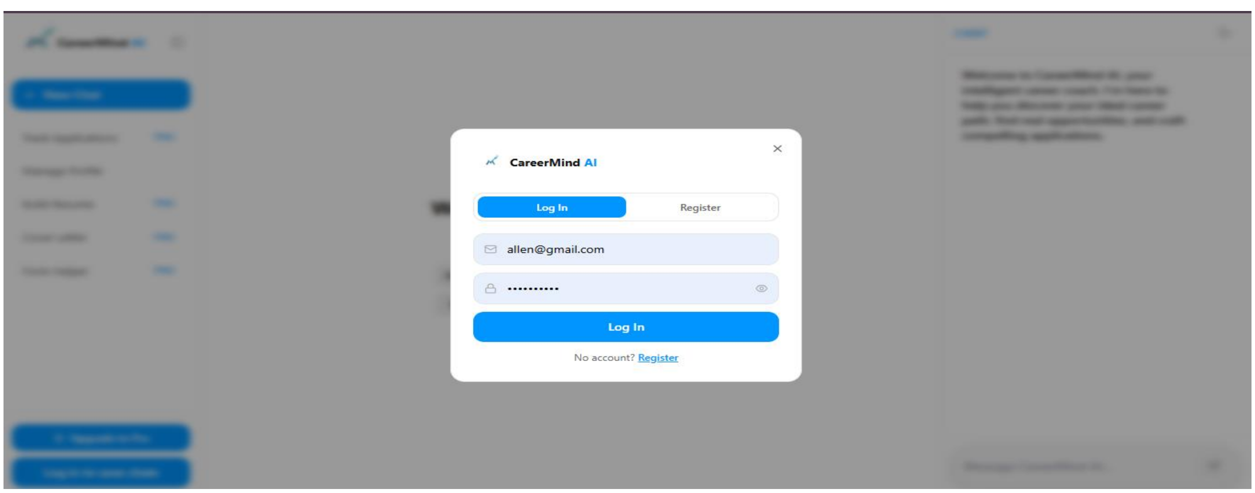
Figure 4.6: CareerMind AI Home Page

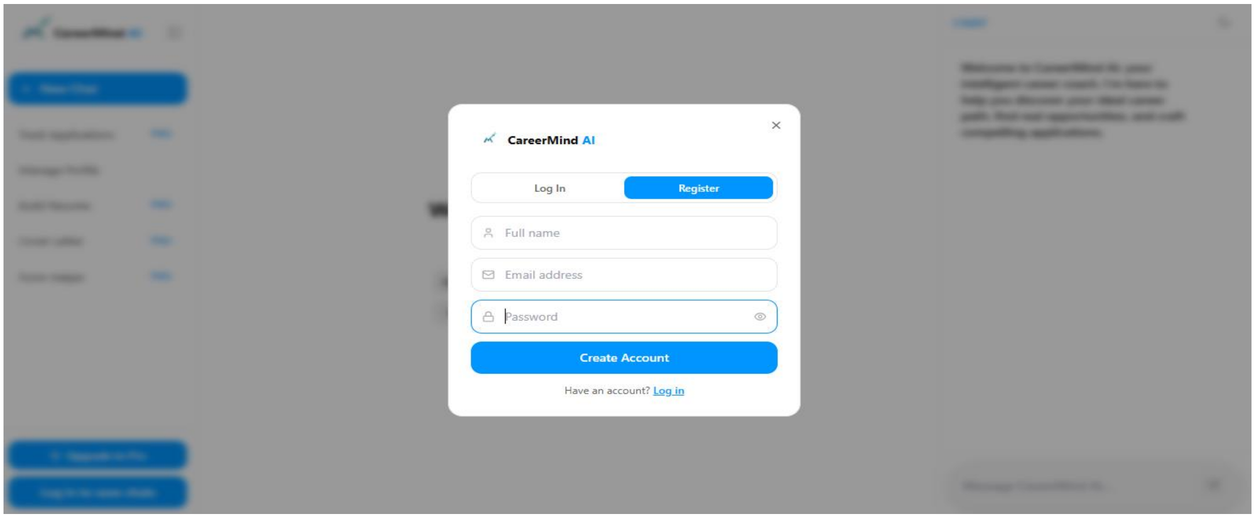


User Registration and Login

Users can create new accounts and securely log into the system to access personalized career guidance services.

Figure 4.7: Login and Registration Page

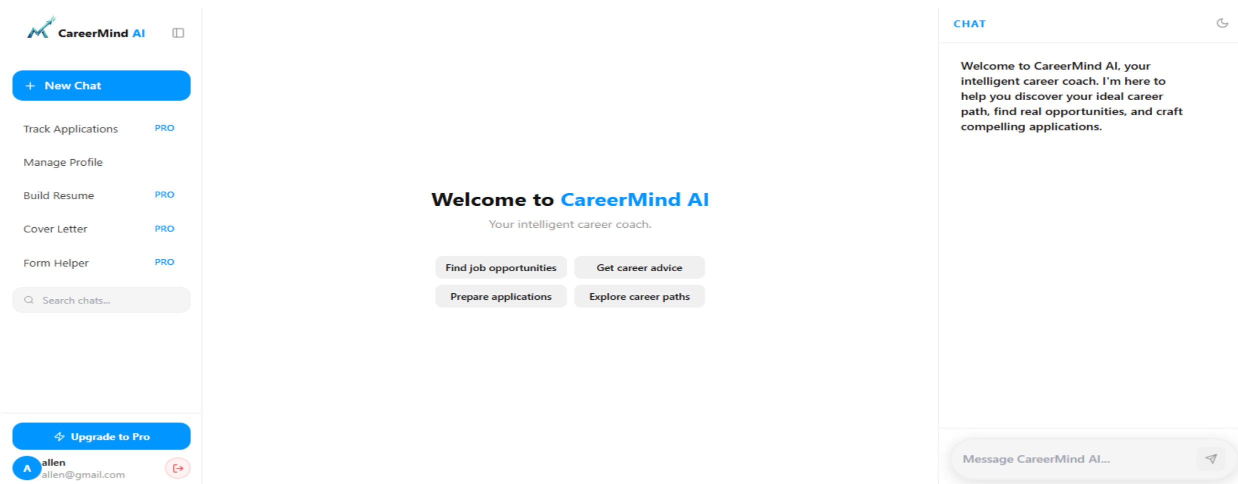




User Dashboard

The dashboard provides access to all major features, including career recommendations, resume generation, cover letter creation, interview preparation, and profile management.

Figure 4.8: User Dashboard

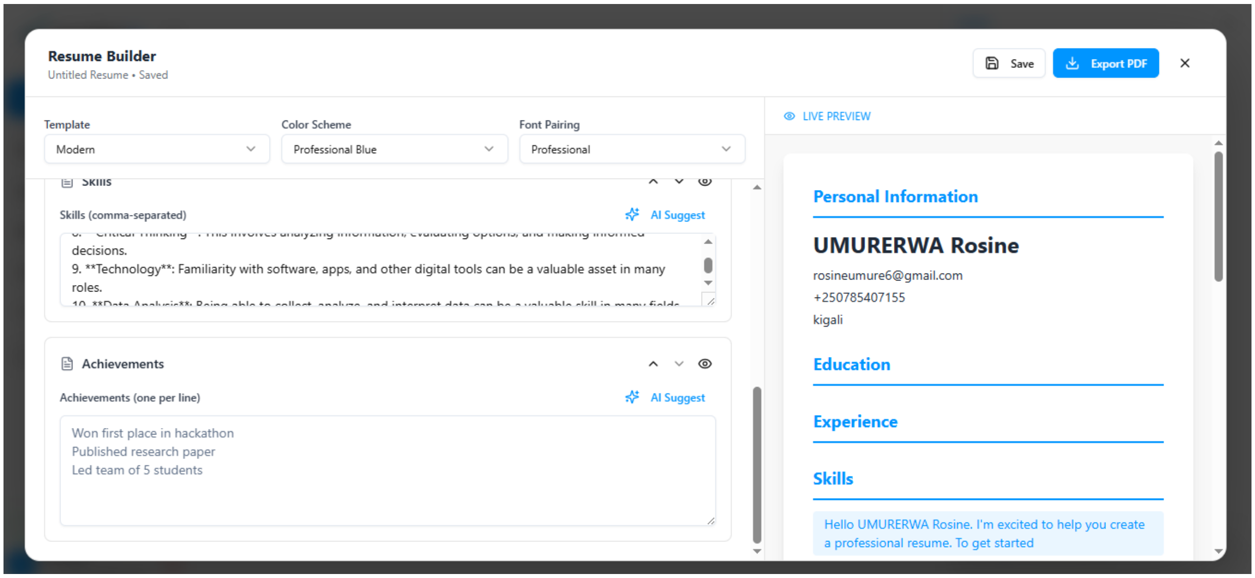


Resume Generator

The Resume Generator enables users to create professional resumes by entering their educational background, skills, experience, and personal information.

Generated resumes can be reviewed before downloading.

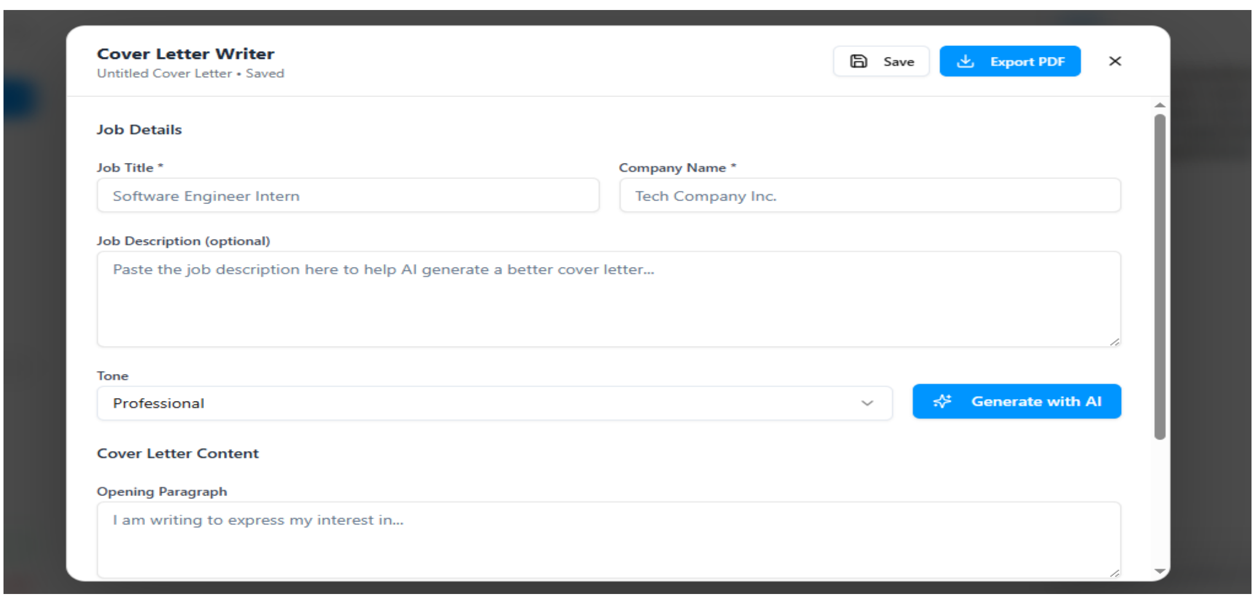
Figure 4.9: Resume Generator



Cover Letter Generator

This module allows users to generate personalized cover letters based on job descriptions and career objectives.

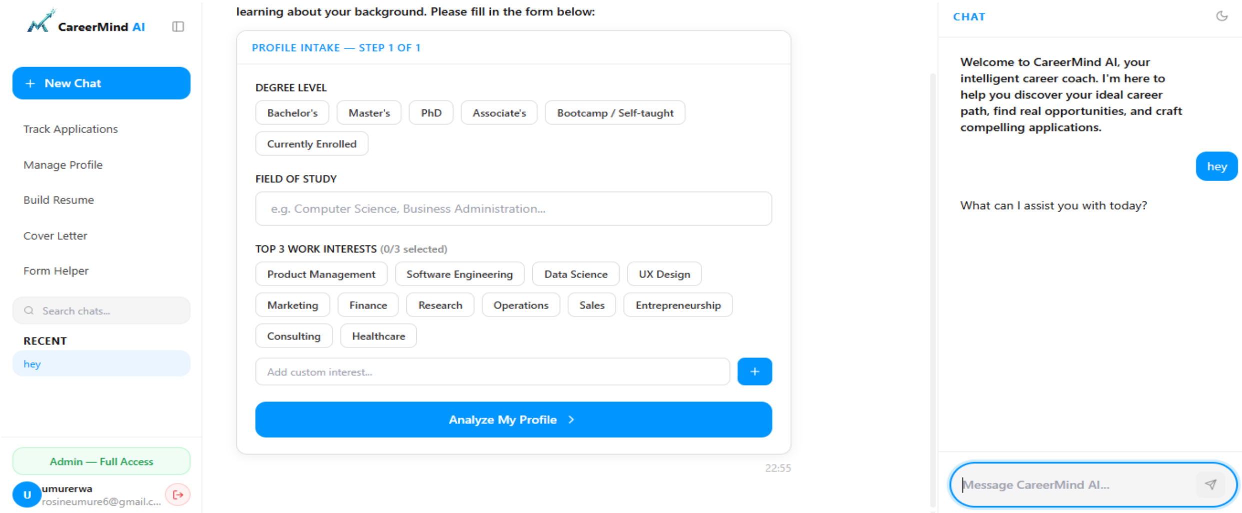
Figure 4.10: Cover Letter Generator



AI Career Assistant

The AI Career Assistant provides personalized career guidance by answering user questions, suggesting career paths, and offering recommendations based on user profiles.

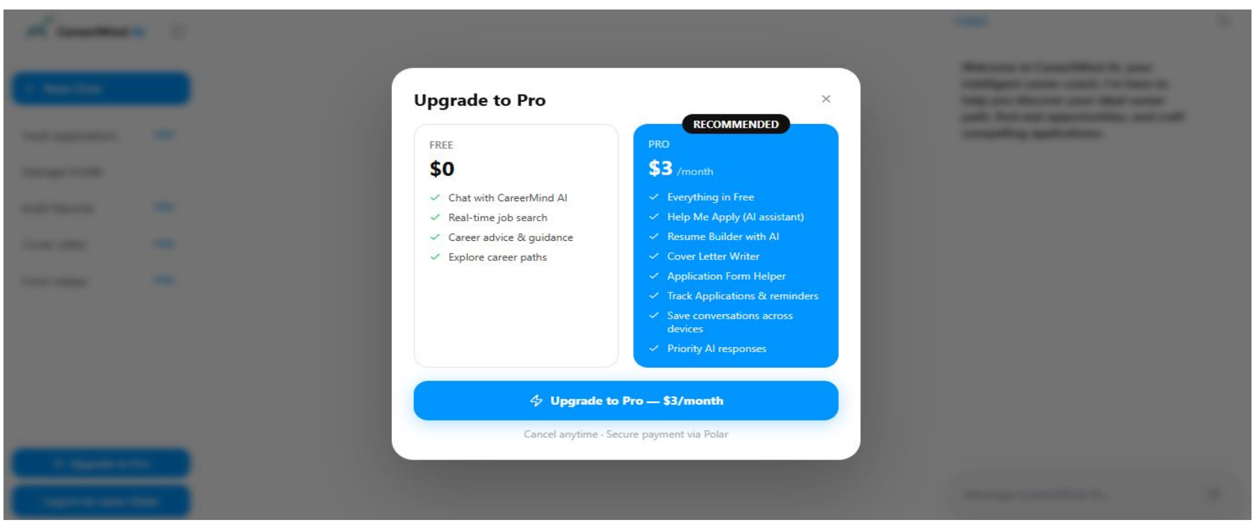
Figure 4.11: AI Career Assistant



Subscription Page

Users can subscribe to premium services for additional AI-powered features and career support.

Figure 4.12: Subscription Page



User Profile

Users can update their personal information, educational background, career interests, and account settings through the profile page.

CONCLUSION

The industrial attachment provided me with valuable practical experience and enabled me to apply the knowledge gained in the classroom to real-world software development projects. Throughout the attachment, I enhanced my technical and professional skills while working on various tasks related to web development, AI, UI/UX design, inventory management, and graphic design.

The major achievement of the attachment was the development of **CareerMind AI**, an AI-powered career guidance platform designed to support students at Rwanda Polytechnic Kigali College. The project demonstrated how modern technologies can be used to solve real-world challenges by providing personalized career guidance, resume and cover letter generation, and interview preparation.

Overall, the attachment strengthened my problem-solving, communication, teamwork, and software development skills. It has prepared me for future professional responsibilities and inspired me to continue developing innovative technology solutions that create a positive impact in society.

CHAPTER FIVE: CHALLENGES, RECOMMENDATIONS, AND CONCLUSION

5.1 Challenges Encountered

During my industrial attachment, I encountered several challenges, including:

- Adapting to new technologies such as React.js, Next.js, Node.js, and Supabase.
- Understanding the structure of large software projects.
- Debugging and fixing software errors during development.
- Managing time while balancing learning and project deadlines.
- Integrating different technologies into one complete application.

Despite these challenges, I overcame them through continuous learning, research, teamwork, and guidance from my supervisors.

5.2 Recommendations

Recommendations to the Organization

- Continue providing interns with hands-on projects to improve practical skills.
- Encourage regular mentorship and knowledge-sharing sessions.
- Provide more opportunities for interns to participate in real-world software development projects.

Recommendations to Rwanda Polytechnic Kigali College

- Strengthen practical training in modern web technologies and Artificial Intelligence.
- Encourage students to work on real-life projects before industrial attachment.
- Continue collaborating with industries to provide quality internship opportunities.

Recommendations for Future Students

- Build a strong foundation in programming before starting industrial attachment.
- Be willing to learn new technologies and ask questions whenever necessary.
- Practice teamwork, time management, and professionalism throughout the attachment.

5.3 Conclusion

The industrial attachment was a valuable learning experience that enabled me to apply theoretical knowledge in a professional environment. Through the development of **CareerMind AI**, I gained practical experience in software development, Artificial Intelligence, and modern web technologies while improving my communication, teamwork, and problem-solving skills. The knowledge and experience gained during this attachment have prepared me for future professional challenges and contributed significantly to my academic and career growth.